

Silicon photomultiplier tube — Solution

Y23 (2023) — English translation

A1

0.40 pts

For steady state, find: voltage V_r at load r ; voltage V_0 on each cell and parasitic capacitance C_0 ; voltage V_1 on the damping layers of each cell.

For steady state, find:

In steady state, no current flows through r , so $V_r = 0$. The voltage on the cells coincides with the source voltage, that is, $V_0 = V_{op}$. The capacitor C_1 is parallel to the resistor, so it is discharged: $V_1 = 0$.

Answer: \textbf{Answer:}

$$V_r = 0$$

$$V_0 = V_{op}$$

$$V_1 = 0$$

A2

0.80 pts

Find: change in charge Δq_0 of capacitor C_0 . Express your answer using C_0 and ΔV ; changes in charges Δq_{1a} and Δq_{2a} of capacitors C_1 and C_2 , respectively, of each of the active cells. Express your answers using C_1 , C_2 , V_{ov} and ΔV ; changes in charges Δq_{1p} and Δq_{2p} of capacitors C_1 and C_2 , respectively, of each of the passive cells. Express your answers using C_{cell} and ΔV .

Find:

By definition of capacity: $\Delta q_0 = C_0 \Delta V$. The charge change on capacitors in active cells is similar: $\Delta q_{1a} = C_1(V_{ov} + \Delta V)$ and $\Delta q_{2a} = -C_2 V_{ov}$, and in passive cells: $\Delta q_{1p} = \Delta q_{2p} = C_{cell} \Delta V$.

Answer: \textbf{Answer:}

$$\Delta q_0 = C_0 \Delta V$$

$$\Delta q_{1a} = C_1(V_{ov} + \Delta V)$$

$$\Delta q_{2a} = -C_2 V_{ov}$$

$$\Delta q_{1p} = \Delta q_{2p} = C_{cell} \Delta V$$

A3

0.70 pts

Find the change in voltage ΔV across the cells and capacitor C_0 upon completion of the avalanche process. Express your answer in general terms through V_{ov} , n , N , C_1 , C_2 , C_0 .

Let us write down the charge conservation law for positive plates C_0 and C_1 .

$$n\Delta q_{1a} + \Delta q_0 + (N - n)\Delta q_{1p} = 0$$

$$nC_1(V_{ov} + \Delta V) + C_0 \Delta V + (N - n)C_{cell} \Delta V = 0$$

$$\Delta V(nC_1 + C_0 + (N - n)C_{cell}) = -nC_1 V_{ov}$$

$$\Delta V = -\frac{nC_1 V_{ov}}{nC_1 + C_0 + (N - n)C_{cell}}$$